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Domain: QA (Manual + Automation)

Assignment 1: Requirement Analysis and User Stories Project: Online Book Store

1. What do customers actually want? (Refining Requirements)

To make this bookstore successful, I think we need to focus on what makes online shopping easy for people today. From my analysis, users don't just want a list of books; they want:

- **Smart Suggestions:** A section that says "If you liked this, you might like..." based on what they've looked at before.
- **Easy Filtering:** People get overwhelmed by too many books, so they need to be able to quickly filter by price, if it's a Hardcover or E-book, and how many stars it has.
- **A Mobile-Friendly View:** Most people shop on their phones during breaks, so the site has to look and work perfectly on a small screen also in addition to other devices (like desktops, tablets, etc.).

2. Functional Requirements (Main Features)

These are the "must-have" parts of the **system**:

- **Accounts:** A way for people to sign up, log in, and save their shipping info so they don't have to type it every time.
- **Search & Browsing:** A search bar at the top for titles or authors, plus a side menu for different genres (like Mystery, Sci-Fi, etc.).
- **The Cart:** A place to hold books before buying, where they can change the quantity or remove things easily.
- **Paying for Books:** We need a secure way to take credit cards or use something like PayPal, so customers feel safe.
- **Tracking Orders:** A simple page where a user can see if their book has shipped yet.

3. Non-Functional Requirements (Behind the Scenes)

These requirements are more about the quality of the user experience rather than specific buttons. We want to make sure the site is reliable and secure enough that customers actually feel comfortable using it.

- **Speed:** No one wants to wait for a page to load. So we need to make sure that the search results to show up in about 2 seconds.
- **Security:** Since we are taking credit card info, we have to use SSL and make sure all user data is encrypted.
- **Reliability:** The site shouldn't crash if 100 people are trying to buy the same bestseller at the same time.

4. User Stories & How They Work (Gherkin Scenarios)

Story A: User Registration

As a new customer,

I want to create an account with my email and a password,

so that my personal information and shipping address are saved for future purchases.

Acceptance Criteria:

- **Given** I am on the "Sign Up" page
- **When** I enter a valid email address and a secure password
- **And** I click the "Create Account" button
- **Then** I should be redirected to my new profile dashboard
- **And** a confirmation email should be sent to my registered address.

Story B: Finding a specific book

As a customer who knows what I want, **I want to** type a book title into a search bar, so that I don't have to scroll through thousands of books to find it.

Acceptance Criteria:

- **Given** I'm on the homepage and see the search bar
- **When** I type in "Atomic Habits" and hit enter

- **Then** the top result should be that specific book
- **And** it should show me right away if it's in stock or sold out.

Story C: Adding things to the cart

As a shopper who likes to browse, **I want to** click an "Add to Cart" button while I'm looking at a book, so that I can keep shopping and pay for everything at the end.

Acceptance Criteria:

- **Given** I am looking at a book's detail page
- **When** I click the "Add to Cart" button
- **Then** a small message should pop up saying it was added
- **And** the little cart icon at the top of the screen should show the number "1".

Story D: Filtering Search Results

As a shopper on a budget,

I want to filter my search results by price range and format (E-book vs. Hardcover),
so that I only see books that I am willing and able to buy.

Acceptance Criteria:

- **Given** I have searched for a genre like "Sci-Fi"
- **When** I select the "E-book" format filter
- **And** I set the price range filter to "Under \$10"
- **Then** the list of books should update to show only items matching both criteria.

Story E: Viewing Order History

As a returning customer,

I want to view a list of my past orders,
so that I can track a current delivery or remember the title of a book I bought previously.

Acceptance Criteria:

- **Given** I am logged into my account
- **When** I click on the "Order History" section of my profile

- **Then** I should see a list of all my previous purchases with dates and totals
- **And** each order should display its current status, such as "Shipped" or "Delivered".

5. Planning the Build (Sprint Plan)

I've decided to split this into two main parts to keep it organized:

- **Sprint 1 (The Basics):** First, we build the account login and the book database. Without these, the store can't really function. We'll also get the search bar working here.
- **Sprint 2 (The Shop & User Experience):** Build the shopping cart, the secure payment gateway, and the filtering system to help users find specific price points.
- **Sprint 3 (Account Management & Post-Purchase):** Develop the user profile dashboard, order history tracking, and the automated "Smart Suggestions" engine.